INCH-POUND

MIL-PRF-28876/9F
w/AMENDMENT 2
5 March 2026

SUPERSEDING
MIL-PRF-28876/9F(SH)
w/INT. AMENDMENT 1
12 October 2023
MIL-PRF-28876/9F
1 March 2021

PERFORMANCE SPECIFICATION SHEET

CONNECTORS, FIBER OPTIC, CIRCULAR, PLUG STYLE, MULTIPLE REMOVABLE TERMINI, SCREW THREADS, WITH 90-DEGREE BACKSHELL, ENVIRONMENT RESISTING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-28876.

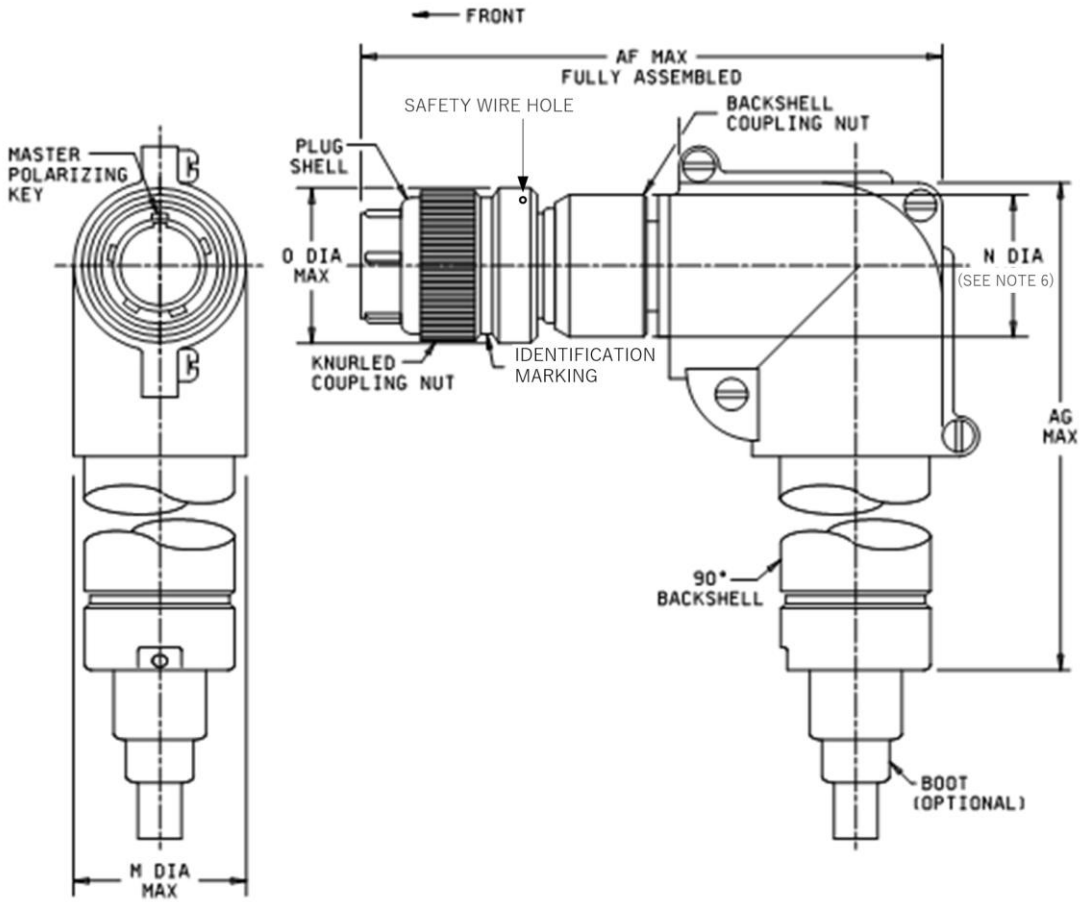


FIGURE 1. Plug connector with 90-degree backshell.

AMSC N/A FSC 6060

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Shell size	M dia max	AF max	AG max	O dia max	Insert figure 1/
11	1.200 (30.48)	4.190 (106.43)	4.250 (107.95)	1.028 (26.11)	B-1
13	1.200 (30.48)	4.190 (106.43)	4.250 (107.95)	1.141 (28.98)	B-2
15	1.550 (39.37)	4.700 (119.38)	4.500 (114.30)	1.263 (32.08)	B-3 or B-4
23	2.250 (57.15)	4.850 (123.19)	5.000 (127.00)	1.705 (43.31)	B-5 or B-6

FOOTNOTE:

1/ For insert arrangement, see the indicated figure in Appendix B of MIL-PRF-28876.

NOTES:

1. Dimensions are in inches.
2. Metric equivalents (mm) are given in parentheses for general information only.
3. Dimensions apply to plated/finished part.
4. Identification marking shall be in accordance with 3.10.3 of MIL-PRF-28876.
5. See appendix A of MIL-PRF-28876 for N DIA dimensioning.
6. Safety wire holes shall be in accordance with MIL-PRF-28876.

FIGURE 1. Plug connector with 90-degree backshell - Continued.

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REQUIREMENTS:

Dimensions and keying configurations. See figure 1 herein and figures A-1, A-3, A-5, A-6, and A-7 of MIL-PRF-28876.

Weight (without termini).

Shell size	Weight, ounces (grams) max
11	9.5 (269.325)
13	11.5 (326.025)
15	14.5 (411.075)
23	27.0 (765.45)

Protective caps. Each connector shall be supplied with protective caps over the front and rear portions of the connector.

Insert arrangement. See Appendix B of MIL-PRF-28876 for desired shell size.

Termini. For style P (pin) terminus or style S (socket) terminus, see MIL-PRF-29504/14 and MIL-PRF-29504/15, respectively. For dummy terminus, see MIL-PRF-29504/3.

Shell polarization. For 1 through 6 keyway positions, see figure A-3 of MIL-PRF-28876.

Part or identifying number (PIN). Marked on the coupling nut of the plug (see table I).

M	**28876**	**/9**	**X**	**X**	**X**	**X**	**X**	**X**
Prefix for military specification	Specification number	Spec sheet no.	Shell size designator (see table I)	Insert arrangement number (see table I)	Backshell dash number (see table I)	Terminus/ insert style (see table I)	Key position number (see table I)	Class designator (see table I)

TABLE I. PIN designators.

Shell size	Shell size designator	Insert arrangement number	Backshell dash number (see table II)	Terminus/ insert style	Key position	Class designator (see table III)
11	A	1	1 or 2	P or S	1, 2, 3, 4, 5, or 6	T or blank
13	B	1	1, 2, or 3	P or S	1, 2, 3, 4, 5, or 6	T or blank
15	C	1 or 2	1, 2, or 3	P or S	1, 2, 3, 4, 5, or 6	T or blank
23	F	1 or 2	1, 2, or 3	P or S	1, 2, 3, 4, 5, or 6	T or blank

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TABLE II. Backshell dash number.

Shell size	Shell size designator	Maximum cable outer diameter in accordance with backshell dash number inch (mm)		
		1	2	3
11	A	0.250 (6.35)	0.346 (8.79)	--
13	B	0.285 (7.24)	0.346 (8.79)	0.453 (11.50)
15	C	0.500 (12.70)	0.250 (6.35)	0.375 (9.53)
23	F	0.866 (22.00)	1.000 (25.40)	0.600 (15.24)

TABLE III. Class designators.

Class designator	Description
Blank	Aluminum with non-conductive finish
T	Aluminum with conductive plating

Example: M28876/9B12P1

Fiber optic cable diameter. See table II.

Mating counterpart. Receptacle connectors are specified in MIL-PRF-28876/1 through MIL-PRF-28876/5 and MIL-PRF-28876/11 through MIL-PRF-28876/14. Dust covers are specified in MIL-PRF-28876/10.

Installation and removal tools. As specified in Drawing 499-6872813.

For qualification, conformance, and periodic inspections, dummy termini shall be used in all unused cavities.

NOTES:

Referenced publications. In addition to MIL-PRF-28876, this specification sheet references the following publications:

MIL-PRF-28876/1	MIL-PRF-28876/2	MIL-PRF-28876/3
MIL-PRF-28876/4	MIL-PRF-28876/5	MIL-PRF-28876/10
MIL-PRF-28876/11	MIL-PRF-28876/12	MIL-PRF-28876/13
MIL-PRF-28876/14	MIL-PRF-29504/3	MIL-PRF-29504/14
MIL-PRF-29504/15	Drawing 499-6872813	

AMENDMENT NOTATIONS. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85
DLA – CC

Preparing activity:
Navy – SH
(Project 6060-2023-024)

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